

# Analysis PhD Exam, May 1997

---

Be sure to write all steps in a neat and logical fashion to receive credit. Give reasons for your steps.

*Definition:* A  $\sigma$ -algebra  $\mathcal{F}$  is separable if  $\mathcal{F} = \sigma\{A_j : j \geq 1\}$  for some sequence of sets  $A_j$ .

1. Let  $\{\mathcal{F}_k\}$  be a sequence of separable  $\sigma$ -algebras of subsets of a set  $X$ . Let  $\mathcal{F}$  be the smallest  $\sigma$ -algebra of subsets of  $X$  which contains  $\mathcal{F}_k$  for each  $k$ . Prove that  $\mathcal{F}$  is separable. [Hint: Use the monotone class theorem.]
2. Let  $(X, \mathcal{F}, \mu)$  be a finite measure space and let  $(Y, \Sigma)$  be a measurable space. Let  $f : X \rightarrow Y$  be  $\mathcal{F}, \Sigma$ -measurable. Define  $\nu(B) = \mu(f^{-1}(B))$  for  $B \in \Sigma$ . Show  $\nu$  is a measure on  $\Sigma$ , and if  $g : Y \rightarrow [0, \infty]$  is  $\Sigma$ -measurable, then  $\int_Y g d\nu = \int_X g \circ f d\mu$ .
3. Let  $m$  be Lebesgue measure on  $\mathbb{R}$ . Let  $f_m = -1_{A_m}$ , where  $A_m = (m, \infty)$ . Show  $(f_m)$  is increasing, but the monotone convergence theorem is not true. Why is this?
4. Let  $(X, \mathcal{S}, \mu)$  and  $(Y, \mathcal{J}, \nu)$  be two finite measure spaces. Construct the product measure  $\mu \times \nu$  on  $\mathcal{S} \otimes \mathcal{J}$ . Be sure to prove  $\mu \times \nu$  is countably additive.
5. Let  $(X, \Sigma, \mu)$  be a finite measure space. Prove that the dual of  $L^1(X, \Sigma, \mu)$  is isometrically isomorphic to  $L^\infty(X, \Sigma, \mu)$ .
6. Let  $(X, \mathcal{F}, \mu)$  be a finite measure space. Suppose that  $\mathcal{F}$  is separable. Show  $L^1(X, \mathcal{F}, \mu)$  is separable.
7. Let  $(X, \mathcal{F}, \mu)$  be a measure space. Let  $\{f_m\}$  be a sequence of measurable functions such that  $\int_X |f_m| d\mu < \frac{1}{2^m}$ . Does it follow that  $\sum_{m=1}^\infty f_m$  converges a.e.  $\mu$  on  $X$ ?
8. Let  $(\Omega, \mathcal{F}, P)$  be a probability space. Let  $X : \Omega \rightarrow \mathbb{R}$  be an integrable function. Let  $F(x) = P(X < x)$ , for  $x \in \mathbb{R}$ .
  - (a) Show  $F : \mathbb{R} \rightarrow [0, 1]$  is monotone, left continuous,  $\lim_{x \rightarrow -\infty} F(x) = 0$ , and  $\lim_{x \rightarrow \infty} F(x) = 1$ .
  - (b) Show  $\int_\Omega X^2 dP = \int_\mathbb{R} x^2 dF(x)$ .
9. Let  $(\Omega, \mathcal{F}, P)$  be a probability space. Let  $(X_m)$  be a sequence of independent random variables on  $\Omega$ .
  - (a) Show  $(\overline{\lim} X_m > 3)$  is a tail event relative to  $(X_m)$ .
  - (b) If  $P(\overline{\lim} X_m > 3) > 0.2$ , what is  $P(\overline{\lim} X_m > 3)$ ?
10. Let  $\ell^\infty$  be the Banach space consisting of all real sequences  $(a_i)$  such that  $\sup_i |a_i| < \infty$ , with  $\|(a_i)\| := \sup_i |a_i|$ . Let  $c$  be the subspace of  $\ell^\infty$  consisting of  $(a_i)$  such that  $\lim a_i$  exists. Prove that there exists a continuous linear functional  $F : \ell^\infty \rightarrow \mathbb{R}$  such that  $F((a_i)) = \lim a_i$  for each  $(a_i) \in c$ . [Hint: Use the Hahn-Banach theorem.]